



Impact of substrate moisture content on growth and metabolic performance of black soldier fly larvae

Nicolai Sundgaard Bekker, Søren Heidelbach, Sofie Zacho Vestergaard, Morten Eneberg Nielsen, Marie Riisgaard-Jensen, Emil Juel Zeuner, Simon Bahrndorff, Niels Thomas Eriksen*

Department of Chemistry and Bioscience, Aalborg University, Fredrik Bajers Vej 7H, 9220 Aalborg, Denmark

ARTICLE INFO

Article history:

Received 29 September 2020

Revised 11 March 2021

Accepted 12 April 2021

Available online 28 April 2021

Keywords:

Hermetia illucens

Respiration

Microbial activity

CO₂

Net growth efficiency

ABSTRACT

Substrate moisture content is an important but not well-understood variable in production and waste reduction processes that involves black soldier fly (BSF) larvae. The purpose of this paper is to characterise growth and metabolic performance of BSF larvae on substrate moisture contents from 45 to 85%. Larvae developed into prepupae only at 45–75% substrate moisture content. Within this interval, the maximal specific growth rate was highest (0.73 day⁻¹), the growth period shortest (13 days), and the maximal dry weight lowest (88 mg) at 45% moisture content. Differences in cost of growth and maintenance were not observed at the different substrate moisture contents, and differences in larval performance were likely associated to differences in co-occurring microbial activities. As much as 22% of the substrate carbon was emitted as CO₂ at 45% moisture content by microorganisms, measured as the difference between total respiration and larval respiration, whereas microbial CO₂ production amounted to only 3% of the substrate carbon at 75% moisture content. As consequence of the high specific growth rate and short growth phase, the overall net growth efficiency was higher at 45% moisture content (0.62) than at 75% moisture content (0.52). Overall, the metabolic performance of the BSF larvae was insensitive to differences in substrate moisture content. Their performance was, however indirectly affected by the substrate moisture content due to differences in co-occurring microbial processes in the substrate.

© 2021 Elsevier Ltd. All rights reserved.

1. Introduction

The black soldier fly (BSF, *Hermetia illucens*) is a promising insect species for upcycling organic waste into high quality biomass (Gold et al., 2020). BSF larvae are able to convert a wide range of organic materials into larval biomass rich in proteins and lipids, and are well suited for use as animal feed (Makkar et al., 2014). Experimental studies on BSF cultivation and waste processing vary considerably in design (Bosch et al., 2019), and basic characterisations of how BSF larvae respond to differences in key rearing conditions may enhance opportunities to compare and evaluate experimental results across studies.

The moisture content in substrates is an important variable in processes involving BSF larvae. Fatchurochim et al. (1989) showed that BSF larvae grow and develop into pupae at substrate moisture contents of 30–70% (based on weight). Other studies suggest that

substrate moisture contents in the range of 50–80% are optimal for larval development, final weight, feed conversion efficiency, and yield (Cheng et al., 2017; Cammack and Tomberlin, 2017; Palma et al., 2018; Abduh et al., 2018; Chen et al., 2019). BSF larvae have even been grown at substrate moisture contents of 90–97.5%, but with lower survival rates and reduced body weights than at 75–88% substrate moisture (Lalander et al., 2020). The substrate moisture content influences substrate texture and affects larval movement, feed consumption, growth and survival (Makkar et al., 2014; Palma et al., 2018). Variations in microbial activity have also been linked to differences in substrate moisture content (Chen et al., 2019). High microbial activities in substrates with low substrate moisture contents may be associated with high porosity and favourable oxygen transfer rates (Palma et al., 2018).

The aim of this study is thus to characterise growth and metabolic performance of BSF larvae at different substrate moisture contents, in terms of specific rates of growth, CO₂ production, and feed assimilation, as well as cost of growth, maintenance, and net growth efficiency (NGE).

* Corresponding author.

E-mail address: nte@bio.aau.dk (N.T. Eriksen).

2. Materials and methods

2.1. Cultivation

BSF eggs were supplied by Entomass Aps (Løkken, Denmark). The eggs (0.2–0.3 g) were placed on top and hatched in nurseries made of cylindrical plastic containers (radius = 5.5 cm, height = 13 cm) containing 100 g (91% dry weight) of a commercial chicken feed (Kyllekræs 3, Danish Agro, Karise, Denmark) and 150 g water. The nurseries were placed at 26.5–28.0 °C and 23–40 %RH. After 9 days, the larvae reached approximately 0.5 mg dry weight (DW) and were named starter larvae (Parodi et al., 2020). Cultivation experiments were carried out in the same containers, each supplemented with 400–600 g of chicken feed (190–225 g DW), and with moisture contents of 45, 55, 65, 75, and 85%, respectively (Table 1). On Day 0 of the cultivation experiments, each container was inoculated with 75–188 starter larvae and 1.2 g of substrate DW per larvae. The amount of substrate and number of starter larvae were adjusted in order to minimise variations between initial substrate mass and substrate height (4.8–6.2 cm), and larval density (0.76–2 cm⁻²). Five replicate cultures were conducted for each treatment. The cultures were placed at 26.5–28.0 °C and 23–40 %RH until the larvae developed into prepupae. Surviving prepupae were counted on Day 19 (28 days after the eggs were placed in the nurseries).

2.2. CO₂ production in BSF larvae

CO₂ production rates were measured on a daily basis on 5 randomly selected larvae from each container. The larvae were washed in 28 °C water to remove substrate residues from their surface and then transferred to an airtight 50 mL plastic centrifugation tube placed in an aluminium heating block (Hettich Benelux model MHR-23) at 28 °C. A pre-calibrated wireless CO₂ Sensor PS-3208 (Pasco, Roseville, California) was inserted into the top of the tube and CO₂ production rates were determined from the linear increase in CO₂ concentration in the tube. Measurements were completed in 5 min, including 2 min for the sensor to stabilise (Laganaro et al., 2021).

2.3. Dry weight content of BSF larvae

After larval CO₂ production rates had been quantified, the wet weight of the 5 randomly selected larvae was measured on an

electronic scale before they were returned alive to their original container. The dry weight of another 3 randomly selected larvae from each container was measured every third day after drying at 95 °C until constant weight. Subsequent estimates of overall respiration rates and mortality did not include these larvae.

2.4. Microbial CO₂ production

The overall CO₂ emission rates from BSF larval culture, including larval as well as microbial CO₂ production, were measured in the headspaces above the substrate using a pre-calibrated wireless CO₂ Sensor PS-3208. The containers were closed by airtight lids during measurements, which were completed in less than 3 min. The CO₂ emission rates were quantified as described in Section 2.2 and microbial CO₂ production was estimated as the difference between the overall CO₂ emission rate and larval CO₂ production rate (calculated as the product of individual CO₂ production rates and the number of larvae in each culture).

2.5. Water holding capacity

The water holding capacity (WHC) of the chicken feed was determined in 5 replicate trials. Chicken feed and water were mixed to a water content above 80% (based on weight) for 24 h in closed containers to ensure water saturation of the chicken feed, and finally transferred to a pre-wetted and pre-weighed coffee filter. Excess water was allowed to drain and the WHC was determined from the increase in mass of drained chicken feed compared to its original dry mass.

2.6. Moisture content in substrate

Controlled amounts of water were added to each container daily, in order to maintain constant substrate moisture contents in the substrates (Supplementary information, Eqs. S1–S4). Firstly, the mass of each container was measured, and the mass of the wet substrate was estimated by subtracting the mass of the BSF larvae. Secondly, it was estimated how much substrate (dry weight) that had been lost via respiration by summing up the measured overall CO₂ production rates and the amounts of substrate incorporated into larval biomass. The remaining substrate dry weight was then estimated as the difference between the initial and metabolised substrate dry weights. The substrate moisture

Table 1

Experimental conditions at Day 0 and experimental and modelled results of larval performance; t_p is the day larvae reached maximal weight, X_{max} is maximal larval biomass in terms of wet weight, dry weight, and carbon content, μ_{max} is maximal specific growth rate, estimated from Eq. (1), and NGE_{avg} is the average net growth efficiency from Day 0 and until the larvae reached maximal weight. Results are mean of 5 replicate cultures, errors indicate standard deviation of replicate measurements.

Parameter/variable	Unit	Experimental conditions Day 0				
Substrate moisture content	%	45	55	65	75	85
Chicken feed	g DW*	220	225	210	150	90
Water	g	158.2	252.7	369.2	435.2	501.1
Substrate moisture content	% WHC**	64	79	93	107	121
Substrate height	cm	4.8	4.9	6.2	5.9	6.2
Number of larvae	–	183	188	175	125	75
Substrate per larva	g DW	1.2	1.2	1.2	1.2	1.2
Areal density	cm ⁻²	1.9	2	1.8	1.3	0.76
Experimental conditions Day 19						
Substrate moisture content	%	43.6 ± 1.1	58.8 ± 1.4	69.4 ± 0.3	79.2 ± 0.3	–
Results						
t_p	day	13	15	15	19	–
X_{max}	mg WW***	299 ± 9	319 ± 12	403 ± 11	388 ± 20	–
	mg DW	84 ± 3	90 ± 3	114 ± 3	104 ± 5	–
	mg C	39.2 ± 0.7	41.9 ± 0.6	52.0 ± 0.6	48.2 ± 0.5	–
μ_{max}	day ⁻¹	0.73 ± 0.05	0.61 ± 0.03	0.61 ± 0.04	0.52 ± 0.02	–
NGE_{avg}	–	0.62	0.57	0.57	0.52	–

*Dry weight, **Water holding capacity, ***Wet weight.

content was then estimated as the difference between the measured substrate wet weight and the estimated substrate dry weight and water was replenished to re-establish the initial substrate moisture content. When cultures were terminated, the moisture content of the remaining substrate in each container was determined after drying at 105 °C until constant weight between 2 subsequent measurements.

2.7. Growth and assimilation of feed

Larval dry weight, X [mg] over time, t was described by the Verhulst logistic growth model from Day 0 after inoculation and until they became prepupae and reached maximal weight, X_{max} (Laganaro et al., 2021)

$$X = \frac{X_{max}}{1 + \left(\frac{X_{max}-X_0}{X_0}\right)e^{-\mu_{max}t}} \quad (1)$$

X_0 is the dry weight of the starter larvae at $t = 0$ and μ_{max} their maximal specific growth rate [day^{-1}]. Eq. (1) was fitted to the measured dry weights of the BSF larvae in order to find the combination of X_{max} and μ_{max} that resulted in the smallest difference (average mean error) between model and data. The specific growth rate, μ of different sized larvae was then found as

$$\mu = \mu_{max} \left(1 - \frac{X}{X_{max}}\right) \quad (2)$$

If growth is balanced, the relationship between specific respiration, q [day^{-1}] and growth rates is expected to be linear (Laganaro et al., 2021).

$$q = Y\mu + m \quad (3)$$

Y and m represent cost of growth (specific amount of substrate metabolised to generate energy for growth) [unitless] and maintenance metabolism [day^{-1}], respectively. The specific feed assimilation rate, a [day^{-1}] was calculated as

$$a = q_{CO_2} + \mu \quad (4)$$

where q_{CO_2} is the specific CO_2 production rate, calculated by dividing CO_2 production rates by larval dry weights. NGE [unit less]

$$NGE = \frac{\mu}{a} = \frac{1}{1 + Y + \frac{m}{\mu}} \quad (5)$$

was estimated from experimental data using the left hand term of Eq. (5). Y and m were estimated by fitting the right hand term (Laganaro et al., 2021) to experimentally determined NGE values. NGE is a measure of an organism's efficiency of converting assimilated feed into biomass (Scriber and Slansky, 1981) and was used as measure of the actual metabolic performance of the BSF larvae during the growth phase. The overall net growth efficiency, NGE_{avg} from Day 0 and until the larvae reached maximal weight was evaluated as the ratio between increases in larval mass divided by the total amount of feed assimilated by the larvae.

2.8. Conversion factors and statistical analysis

In order to compare measurements of growth and CO_2 production, measurements were expressed as specific rates based on the mass of carbon in the larvae, in substrates, and in CO_2 . We assumed that carbon made up 49% of the organic fraction of the larvae, which is the carbon content in biomass of average elementary composition (Roels, 1980). The organic fraction of the larvae was estimated from measurements of the dry weight of the larvae, assuming 5% ash content (Laganaro et al., 2021). We also assumed that the carbon content of chicken feed was 49% of the organic fraction (94.8% of the dry weight).

Maximal weights and maximal specific growth rates of BSF larvae grown at different substrate moisture contents were compared by a Tukey test on results from one-way ANOVA in Excel 2016. A preliminary study had verified that individual larval dry weight followed Gaussian distributions throughout the growth phase.

3. Results and discussion

Original data are available at Mendeley Data (<https://doi.org/10.17632/k435zc736t.1>).

3.1. Cultivation conditions

The WHC of the chicken feed was $70 \pm 0.2\%$ ($n = 5$). Substrate moisture contents of 45–85% thus corresponded to 64–121%, respectively of the WHC (Table 1) and affected culture conditions. In between daily samplings, clumps developed at the surface of substrates with 45% moisture content and most larvae were observed in the bottom half. Substrates with 55% and 65% moisture contents developed a crust at the surface and larvae were found at all depths. Substrates with 75% moisture content had a thin water film at the surface and larvae were predominantly observed in the upper half. Substrates with 85% moisture content had a water phase at the surface and all larvae died. BSF larvae have survived higher moisture contents in other substrates (Lalander et al., 2020), maybe due to differences in WHC. Since the substrates became stratified, substrate moisture contents could not be assayed from sampling during the growth period. Instead, water was replenished based on a theoretical approach (Supplementary information, Fig. S1). The substrate moisture content measured on Day 19, when the cultures were terminated, had remained close to the expected levels (Table 1).

The BSF larvae grew and developed into prepupae in substrates with 45–75% moisture content (Fig. 1A, C, E, G) and reached 299–403 mg wet weight (WW, Table 1). Prepupae commonly weighs 55–299 mg WW depending on substrate (see e.g. Gold et al., 2018; Lalander et al., 2019) but weights up to at least 373 mg have been seen (Lalander et al., 2020). The high weight of the prepupae in this study indicates that suitable growth conditions had been established. On Day 19, when cultures were harvested, 91 ± 4 , 89 ± 4 , 95 ± 10 , and $81 \pm 11\%$ of the larvae were alive as prepupae at 45, 55, 65, and 75% moisture content, respectively. Maximal weights were reached on Day 13–19, earliest in the driest substrates (Table 1). Fatchurochim et al. (1989) also observed that BSF larvae grew for longer periods in wet compared to more dry substrates. Growth followed sigmoidal curves with exponential phases of 8–12 days (Supplementary information, Fig. S2). The larvae grew larger at 65% and 75% substrate moisture content (104–114 mg DW) than at 45% and 55% substrate moisture content (84–90 mg DW, $p = 2.7 \times 10^{-9}$). The relationship between dry weight content and larval size is shown in Supplementary information, Fig. S3). The maximal specific growth rate was, in contrast, higher at 45% (0.73 day^{-1}) compared to 75% substrate moisture content (0.52 day^{-1}), $p = 2.8 \times 10^{-6}$. These values fall within the wide interval of maximal specific growth rates on different substrates (0.05 – 1.08 day^{-1}) that previously have been estimated by fitting sigmoidal curves to larval body widths (Sripontan et al., 2020). Specific growth rates estimated from ln-transformed larval weights confirmed that highest value occurred at lowest substrate moisture content (Supplementary information, Fig. S2). Highest specific growth rates were thereby found at substrate moisture contents below levels that in other studies have favoured growth and development of BSF larvae (Cheng et al., 2017; Cammack and Tomberlin, 2017; Palma et al., 2018; Chen et al., 2019).

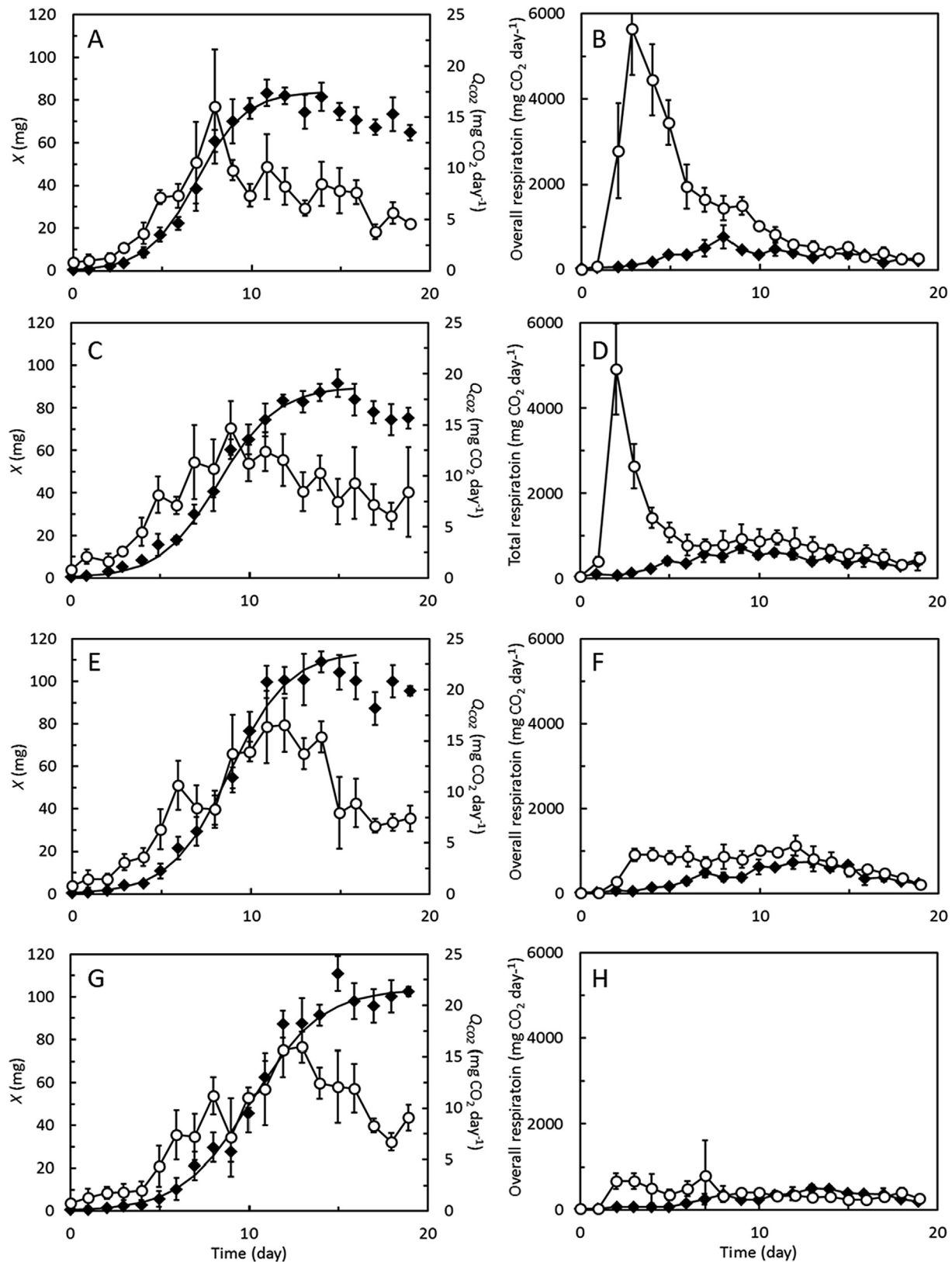


Fig. 1. BSF larvae grown at substrate moisture contents of 45% (A-B), 55% (C-D), 65% (E-F), and 75% (G-H). A, C, E, G. Larval dry weight X (◆, curve predicted from Eq. (1)), individual CO_2 production rate Q_{CO_2} (○). B, D, F, H. Overall CO_2 emission rate from cultures (○) and from larvae (◆). Data are averages of 5 replicate cultures $\pm$ standard deviation.

3.2. Microbial activities

The substrate moisture content strongly affected the microbial CO₂ production in the substrates (Fig. 1B, D, F, H). At 45% and 55% substrate moisture contents, the overall CO₂ emission vastly exceeded larval CO₂ production (Table 2). Microbial processes converted 12–22% of the substrate carbon into CO₂, thereby making this fraction unavailable for the BSF larvae. At 65% and 75% substrate moisture content, the BSF larvae dominated the CO₂ emissions. Microbial CO₂ emissions peaked already after 3–4 days at 45–55% substrate moisture content. The lower microbial CO₂ emission rates at the end of the cultivations (Fig. 1B and D) suggest that the most readily available feed compounds had been consumed within the first week. Similarly during composting processes the soluble and most available fractions may be degraded within less than a week, giving rise to decreasing oxygen uptake rates (Xi et al., 2005). Thus, microbial activities therefore affected the feed value of the remaining substrate in late growth phases. Parodi et al. (2020) also observed a peak in CO₂ emissions 4 days after BSF cultures had been started, where after the emission rates decreased. The shorter growth phase and lower maximal weight of prepupae in the driest substrates (Table 1) were likely results of microbial deterioration of substrate availability and quality. It should be noted that CO₂ emissions into culture headspaces represent only a fraction of the microbial activities. Some CO₂ may remain dissolved in pore water and microorganisms may convert substrate components into volatile fermentation products. Indeed, culture mass balances from Day 0 to Day 19 (Supplementary information, Table S1) confirmed that the CO₂ measurements did not fully explain the overall decrease in substrate weight after drying. Cumulative CO₂ emissions and larval growth summed up to 87% of the removed substrate at 45% substrate moisture content and only 45% at 75% substrate moisture content. A similar result (just 47–75% carbon removal via CO₂ and larval biomass) can be deduced from Pang et al. (2020), who found that 5–17% of the carbon initially present in substrates made from pig manure and corncob was lost via routes different from CO₂ (9–12%), CH₄ (less than 0.2%), and larval biomass (4–7%).

3.3. Metabolic performance

The carbon content in the BSF larvae, the cumulative amount of carbon emitted as CO₂, and the accumulated amount of carbon assimilated by the larvae is shown in Fig. 2A, C, E, G. At the time the larvae reached maximal weight, corresponding to 39–52 mg C (Table 1), each larva had assimilated feed corresponding to 72–91 mg C, resulting in NGE_{avg} between 0.52 and 0.62 (Table 1). Highest NGE_{avg} was in the larvae reared at lowest substrate moisture content. Gold et al. (2020) found that carbon made up 62% of the organic fraction of BSF prepupae. Such high level will hardly be reached in growing larvae due to their lower lipid contents, but if so, NGE_{avg} estimates would have been slightly higher (0.56–0.65) compared with our results using 49% carbon content. In all circumstances, the high NGE_{avg} at low substrate moisture contents was a consequence of the high specific growth rates and short

growth phases (Table 1) since the larvae spend more substrate for maintenance purposes the longer the growth phase was.

Specific rates of growth, CO₂ production, and feed assimilation were highest in the smallest larvae and decreased as the larvae approached maximal weight (Fig. 2A, C, E, G). The specific growth rate and thus the specific feed assimilation rate reached highest values at 45% substrate moisture content. From Day 2, when the larvae had grown larger than 1 mg DW, and until growth stopped, the specific CO₂ production rate became linearly correlated to the specific growth rate at 45% and 55% substrate moisture content (Fig. 2B and D), as predicted by Eq. (3). At 65% and 75% substrate moisture content, the relationship between the specific rates of CO₂ production and growth became linear only after Day 4 and Day 5, when the larvae were larger than 5 and 6 mg DW, respectively. Until then, measured specific CO₂ production rates were inexplicable high and we cannot rule out that respiration measurements on the smallest larvae may have been affected by so-called ‘flask effects’ (Wightman, 1981). These larvae are fragile and the separation of larvae from substrates during respiration measurements may have been stressful. However, only minor parts of the biomass (1–6%) were produced these first days of growth and the estimated NGE_{avg} values (Table 1) can potentially have been affected only little.

The cost of growth, $Y = 0.38$ and maintenance, $m = 0.05 \text{ day}^{-1}$ were found by simultaneously fitting Eq. (5) to the estimated NGE values in all cultures, during the parts of the growth phases where the specific CO₂ production rate depended linearly on the specific growth rate (Fig. 2B, D, F, H). More than 90% of the biomass was produced during these phases. These values described well the linear relationships between the specific rates of CO₂ production and growth (Eq. (3)) as well as the non-linear relationships between NGE and specific growth rate at all substrate moisture contents. Similar cost of growth and maintenance were also found by Laganaro et al. (2021). Thereby cost of growth and maintenance metabolism seemed not to be affected by substrate moisture content, maybe with the exception of larvae smaller than 1–6 mg DW. Key processes, which affected larval performances in this study are depicted in Supplementary information, Fig. S4 and did include larval as well as microbial substrate utilisation. Particularly the latter were affected by substrate moisture content.

4. Conclusion

The results presented in this paper suggest that the metabolic performance in terms of cost of growth and maintenance in BSF larvae is insensitive to differences in substrate moisture content within a broad interval from 45 to 75% (64–107% of the WHC). This however, do not mean that BSF larvae will perform the same at all substrate moisture contents. In this study, low substrate moisture contents favoured microbial substrate degradation to CO₂, resulting in comparatively short growth phases and small prepupae but efficient conversion of assimilated substrate into larval biomass. The latter was also strengthened by the high specific growth rates observed at low substrate moisture contents. The insensitivity of BSF larvae to differences in substrate moisture content may

Table 2

Initial carbon contents in substrates, Day 0 compared to cumulative overall CO₂ emissions, cumulative larval CO₂ productions, and cumulative microbial CO₂ productions. Values are mean of 5 replicate cultures $\pm$ standard deviation.

	Unit				
Substrate moisture content	%	45	55	65	75
Substrate carbon content, Day 0	g C	102.2	104.5	97.5	69.7
Overall CO ₂ emission	g C	28.2 $\pm$ 1.4	20.4 $\pm$ 2.7	13.3 $\pm$ 0.7	7.0 $\pm$ 1.3
Larval CO ₂ production	g C	6.2 $\pm$ 0.6	7.6 $\pm$ 0.7	7.5 $\pm$ 0.6	5.0 $\pm$ 0.6
Microbial CO ₂ production	g C	22.0 $\pm$ 1.0	12.7 $\pm$ 2.5	5.8 $\pm$ 0.8	2.0 $\pm$ 1.6

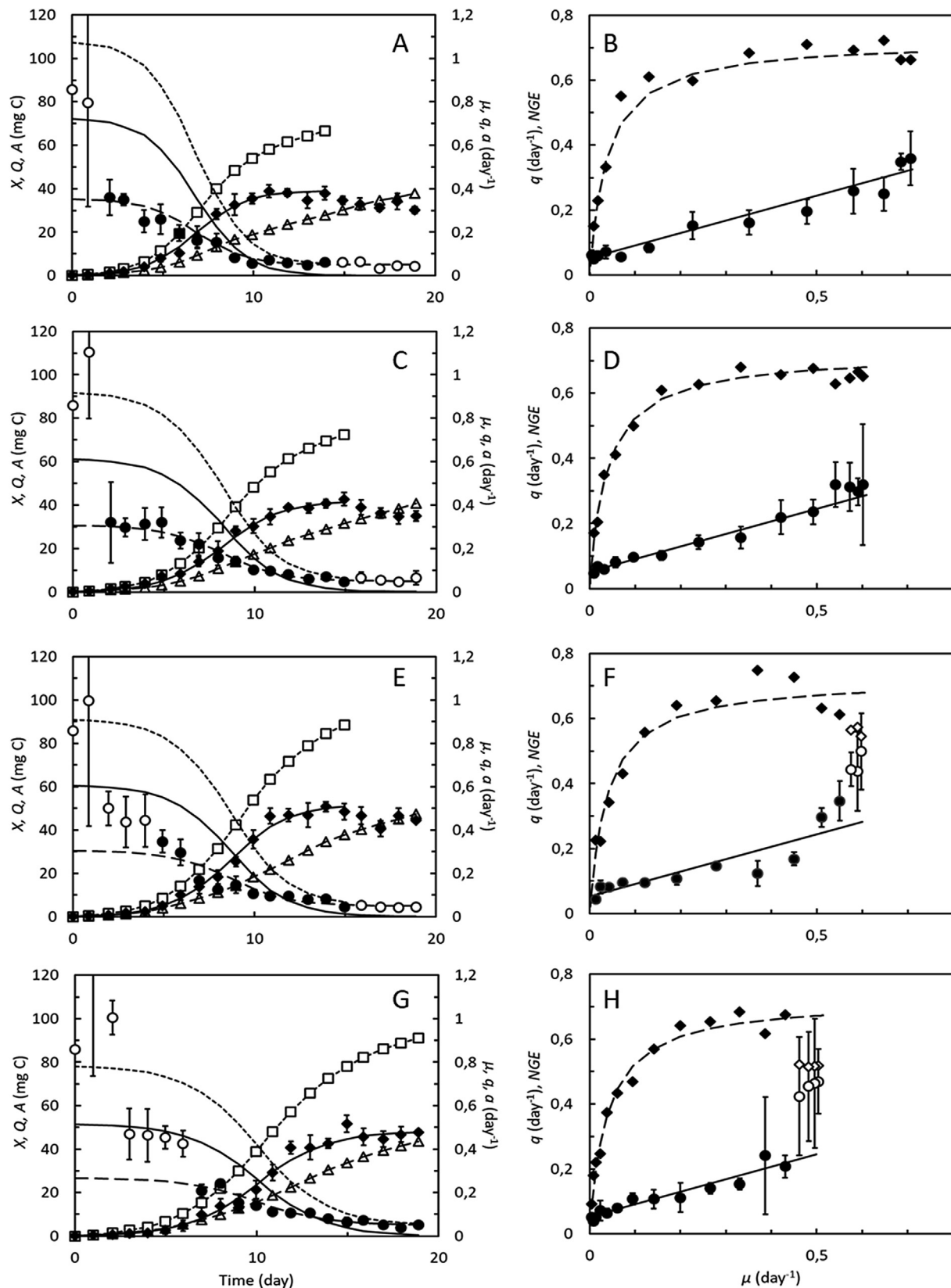


Fig. 2. BSF larvae grown at substrate moisture contents of 45% (A-B), 55% (C-D), 65% (E-F), and 75% (G-H). A, C, E, G. Larval carbon content, X (●, curve predicted from Eq. (1)), cumulative CO₂ production, Q (Δ), and cumulative amount of carbon assimilated by the larvae, A (□), specific growth rate, μ (solid curve predicted from Eq. (2)), specific CO₂ production rate, q_{CO_2} (○, ●, curve predicted from Eq. (3)), solid symbols represent measurements in line with model predictions), specific assimilation rate, a (dotted curve predicted from Eq. (4)). B, D, F, H. Specific CO₂ production rate, q_{CO_2} (○, ●, solid line) and net growth efficiency, NGE (◇, ◆, dashed curve) vs. specific growth rate, μ . Curves are predicted from Eqs. 3–5 with $Y = 0.38$ and $m = 0.05$ day⁻¹. Solid symbols represent data points in line with model predictions. Data are averages of 5 replicate cultures $\pm$ standard deviation.

provide freedom with respect to process design and substrate formulation and is advantageous for cross study comparisons of experimental results. The sensitivity of microbial processes to substrate moisture content and the resulting effects on larval performances do, on the other hand, emphasise that substrate moisture content is a key variable with respect to the overall performance of processes involving BSF larvae.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

We thank the company Entomass Aps for cooperation and supply of BSF eggs, and Helle Blendstrup for valuable assistance in the laboratory.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.wasman.2021.04.028>.

References

- Abduh, M.Y., Nadiia, M.H., Syaripudina, Manurunga, R., Putraa, R.E. 2018. Factors affecting the bioconversion of Philippine tung seed by black soldier fly larvae for the production of protein and oil-rich biomass. *J. Asia-Pacific Entomol.* 21, 836–842.
- Bosch, G., van Zanten, H.H.E., Zamprognia, A., Veenenbos, M., Meijer, N.P., van der Fels-Klerx, H.J., van Loon, J.J.A., 2019. Conversion of organic resources by black soldier fly larvae: Legislation, efficiency and environmental impact. *J. Cleaner Prod.* 222, 355–363.
- Cammack, J.A., Tomberlin, J.K., 2017. The impact of diet protein and carbohydrate on select life-history traits of the black soldier fly *Hermetia illucens* (L.) (Diptera: Stratiomyidae). *Insects* 8, 1–14.
- Chen, J., Hou, D., Pang, W., Nowar, E.E., Tomberlin, J.K., Hu, R., Chen, H., Xie, J., Zhang, J., Yu, Z., Li, Qing, 2019. Effect of moisture content on greenhouse gas and NH₃ emissions from pig manure converted by black soldier fly. *Sci. Tot. Environ.* 697, 133–840.
- Cheng, J.Y.K., Chiu, S.L.H., Lo, I.M.C., 2017. Effects of moisture content of food waste on residue separation, larval growth and larval survival in black soldier fly bioconversion. *Waste Manage.* 67, 315–323.
- Fatchurochim, S., Geden, C.J., Axtell, R.C., 1989. Filth fly (Diptera) oviposition and larval development in poultry manure of various moisture levels. *J. Entomol. Sci. Georgia Entomol. Soc.* 24, 224–231.
- Gold, M., Cassar, C.M., Zurbrugg, C., Kreuzer, M., Boulos, S., Diener, S., Mathys, A., 2020. Biowaste treatment with black soldier fly larvae: Increasing performance through the formulation of biowastes based on protein and carbohydrates. *Waste Manage.* 102, 319–329.
- Gold, M., Tomberlin, J.K., Diener, S., Zurbrugg, C., Mathys, A., 2018. Decomposition of biowaste macronutrients, microbes, and chemicals in black soldier fly larval treatment: A review. *Waste Manage.* 82, 302–318.
- Laganaro, M., Bahrndorff, S., Eriksen, N.T., 2021. Growth and metabolic performance of black soldier fly larvae grown on low and high-quality substrates. *Waste Man.* 121, 198–205.
- Lalander, C., Diener, S., Zurbrugg, C., Vinnerås, B., 2019. Effects of feedstock on larval development and process efficiency in waste treatment with black soldier fly (*Hermetia illucens*). *J. Cleaner Prod.* 208, 211–219.
- Lalander, C., Ermolaev, E., Wiklicky, V., Vinnerås, B., 2020. Process efficiency and ventilation requirement in black soldier fly larvae composting of substrates with high water content. *Sci. Tot. Environ.* 729, 138968.
- Makkar, H.P.S., Tran, G., Heuzé, V., Ankers, P., 2014. State-of-the-art on use of insects as animal feed. *Animal Feed Sci.* 197, 1–33.
- Palma, L., Ceballos, S.J., Johnson, P.C., Niemeier, D., Pitesky, M., VanderGheynst, J.S., 2018. Cultivation of black soldier fly larvae on almond byproducts: impacts of aeration and moisture on larvae growth and composition. *J. Sci. Food Agricul.* 98, 5893–5900.
- Pang, W., Hou, D., Nowar, E.E., Chen, H., Zhang, J., Zhang, G., Li, Q., Wang, S., 2020. The influence on carbon, nitrogen recycling, and greenhouse gas emissions under different C/N ratios by black soldier fly. *Environ. Sci. Pollut. Res.* 27, 42767–42777.
- Parodi, A., De Boer, I.J.M., Gerrits, W.J.J., Van Loon, J.J.A., Heetkamp, M.J.W., Van Schelt, J., Bolhuis, E., Van Zanten, H.H.E., 2020. Bioconversion efficiencies, greenhouse gas and ammonia emissions during black soldier fly rearing - A mass balance approach. *J. Cleaner Produc.* 271, 122488.
- Roels, J.A., 1980. Application of macroscopic principles to microbial metabolism. *Biotechnol. Bioeng.* 22, 2457–2514.
- Scriber, J.M., Slansky, F., 1981. The nutritional ecology of immature insects. *Ann. Rev. Entomol. Ann. Rev.* 26, 183–211.
- Sripontan, Y., Chiu, C.-I., Tanansathaporn, S., Leasen, K., Manlong, K., 2020. Modeling the growth of black soldier fly *Hermetia illucens* (Diptera: Stratiomyidae): An approach to evaluate diet quality. *J. Econom. Entomol.* 113, 742–751.
- Wightman, J.A., 1981. Why insect energy budgets do not balance. *Oecologia* 50, 166–169.
- Xi, B., Zhang, B., Liu, H., 2005. Process kinetics of inoculation composting of municipal solid waste. *J. Hazard. Mat. B* 124 (2005), 165–172.